

Phase 4: Applying Machine Learning Methods

Music Listening Behavior & Weather Analysis
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1. Introduction

In this phase, four supervised machine learning methods are applied to the daily aggregated dataset (music_weather_daily.csv, 900 observations, Oct 2023 to present) to model the relationship between Istanbul weather and personal Apple Music listening behavior. Methods 1 and 3 ask whether weather predicts listening behavior. Methods 2 and 4 reverse the question — testing whether listening behavior can predict weather. This symmetric design tests whether any relationship holds in both directions.

#	Method	Task	Target	Features
1	Linear Regression	Regression	total_listen_min	Weather features
2	Logistic Regression	Classification	is_rainy	Listening features
3	KNN	Regression	total_listen_min	Weather + time + listening
4	Decision Tree	Classification	condition (4-class)	Listening + time features

2. Linear Regression

Linear regression is applied as the baseline model. The question asked is: can daily weather conditions linearly predict how many minutes I listen to music?

Features: temp_mean, temp_max, precipitation, sunshine_hours **Target:** total_listen_min

Results:

Metric	Value
R ² Score	0.0499
MAE	59.20 minutes

Feature	Coefficient
temp_mean	-4.587
temp_max	+1.884
precipitation	-0.297
sunshine_hours	+0.565

Linear Regression: Weather → Listening Duration

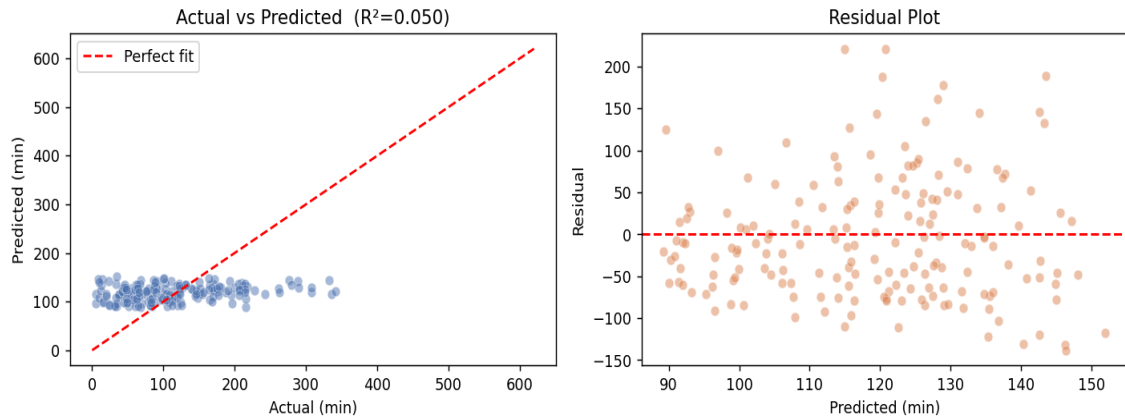


Figure 1: Actual vs Predicted and Residual Plot for Linear Regression

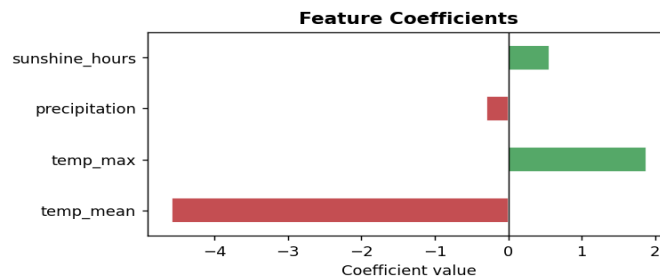


Figure 2: Feature Coefficients

The model achieves $R^2 = 0.05$ and MAE = 59 minutes, meaning weather features alone explain almost none of the variance in daily listening time. The residual plot shows large, uniform scatter with no systematic pattern — the model is not misspecified, there are simply important variables missing. Notably, precipitation has a negative coefficient, contradicting the intuitive assumption that rainy days lead to more listening. This is consistent with Phase 3 hypothesis testing results.

Key takeaway: Weather does not linearly predict listening time. This motivates more flexible methods.

3. Logistic Regression

Logistic regression reverses the direction — can listening behavior predict whether it was a rainy day? Features are standardised using StandardScaler since logistic regression is scale-sensitive.

Features: n_plays, total_listen_min, skip_rate, fraction_complete **Target:** is_rainy

Metric	Value
Accuracy	0.7056
Majority Baseline	0.7056
Lift over Baseline	0.0000

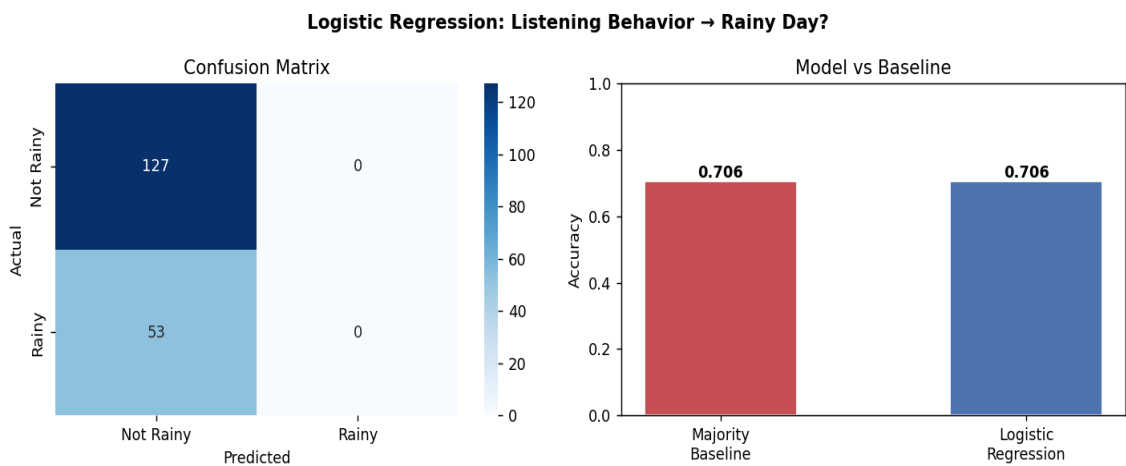


Figure 3: Confusion Matrix and Accuracy vs Baseline for Logistic Regression

The model achieves 70.6% accuracy, identical to the majority-class baseline (predict "not rainy" every day). The confusion matrix confirms this: recall for rainy days is effectively zero. This is a direct consequence of class imbalance (72% of days are non-rainy) combined with the absence of any discriminative signal in listening features. This mirrors the weak correlations found in Phase 3 hypothesis testing.

Key takeaway: Listening patterns carry no discriminative signal about rain. The relationship is not symmetric.

4. K-Nearest Neighbors (KNN) Regression

KNN regression predicts listening time by averaging the k most similar historical days. A richer feature set is used compared to linear regression, and features are standardised since KNN is distance-based. The hyperparameter k is tuned across 6 values.

Features: temp_mean, precipitation, sunshine_hours, n_plays, skip_rate, day_of_week, month
Target: total_listen_min

k	R ² Score	MAE (min)
1	0.6478	31.78
3	0.7288	29.92
5	0.7569	27.74
10	0.7653	26.87

20	0.7286	28.93
50	0.6766	32.51

KNN: k Tuning Results

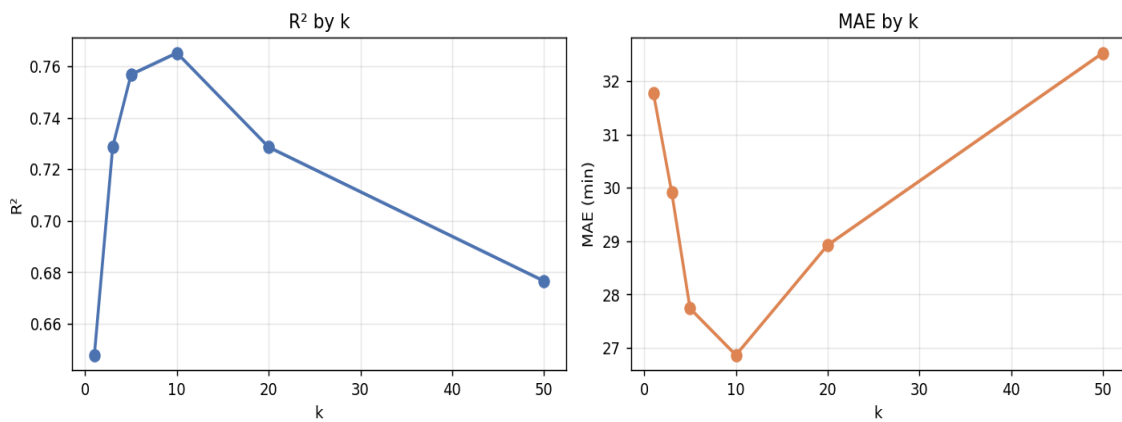


Figure 4: R^2 and MAE across k values

KNN (k=10): All Features → Listening Duration

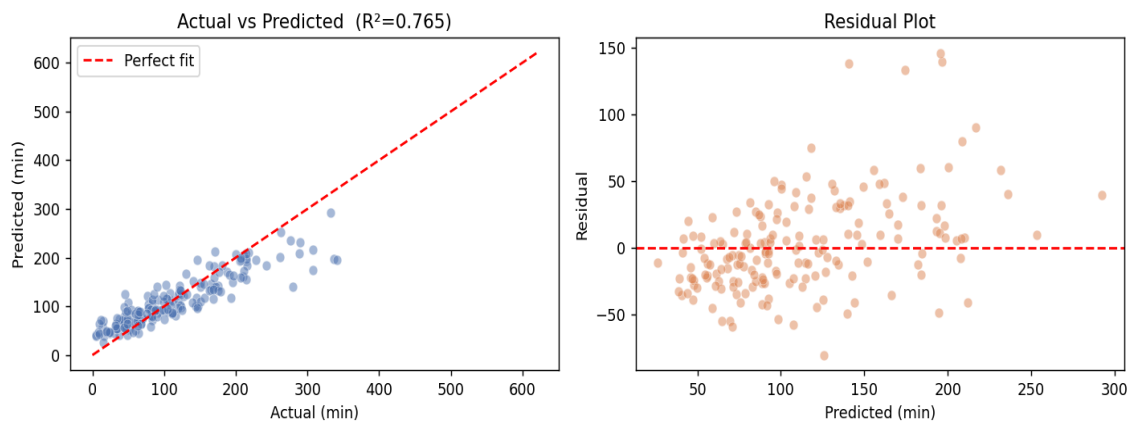


Figure 5: Actual vs Predicted and Residual Plot for Best KNN ($k=10$)

Best KNN ($k=10$): $R^2 = 0.77$, MAE = 27 minutes — a dramatic improvement over linear regression ($R^2 = 0.05$). The k -tuning curve shows a clear sweet spot at $k=10$: $k=1$ overfits the training data, while $k=50$ over-smooths and loses local structure. The improvement from 0.05 to 0.77 comes from including time and behavior features beyond weather, and allowing nonlinear relationships.

Key takeaway: Listening duration is driven by complex nonlinear interactions. Weather alone is insufficient.

5. Decision Tree Classification

A decision tree is applied to test whether weather condition (Rainy, Partly Cloudy, Clear, Snowy) can be inferred from listening behavior and time features alone — deliberately excluding all weather variables. Max depth is tuned to prevent overfitting.

Features: n_plays, total_listen_min, skip_rate, fraction_complete, day_of_week, month, year (no weather) **Target:** condition

max_depth	Accuracy
2	0.4833
3	0.4944
4	0.4778
5	0.4556
7	0.4722
10	0.4444
None	0.4056

Decision Tree: Listening+Time → Weather Condition

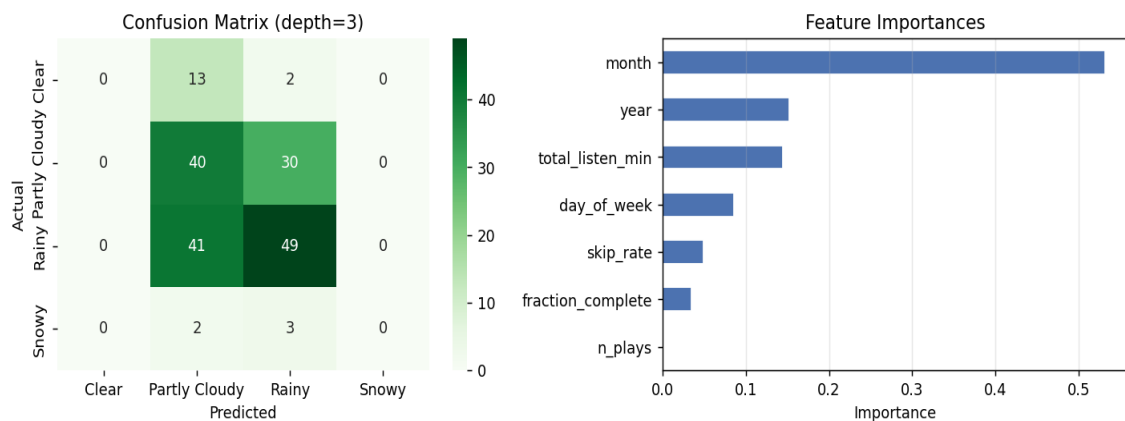


Figure 6: Confusion Matrix and Feature Importances for Decision Tree (depth=3)

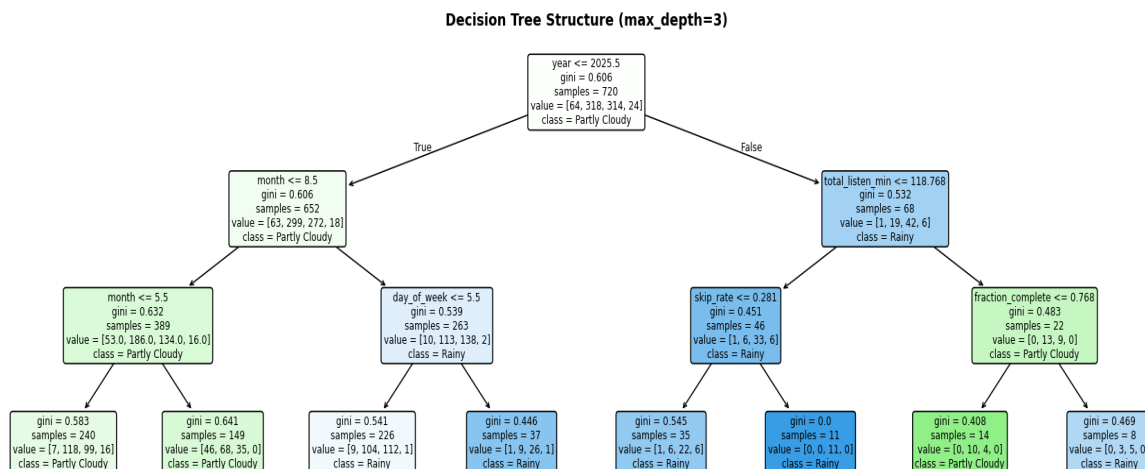


Figure 7: Decision Tree Structure — First 3 Levels

Best decision tree (depth=3) achieves 49.4% accuracy on a 4-class problem with a majority baseline of ~50%. Month dominates feature importance at 53% — weather condition is primarily encoded by time of

year, not listening patterns. Pure listening features (n_plays, skip_rate) contribute minimally, confirming the asymmetric and limited nature of the weather-music relationship.

Key takeaway: Time features encode weather far better than listening patterns do.

6. Model Comparison & Summary

Model	Task	Metric	Score
Linear Regression	Regression	R ²	0.0499
Linear Regression	Regression	MAE (min)	59.20
Logistic Regression	Classification	Accuracy	0.7056
KNN (k=10)	Regression	R ²	0.7653
KNN (k=10)	Regression	MAE (min)	26.87
Decision Tree (d=3)	Classification	Accuracy	0.4944

Phase 4: Model Performance Summary

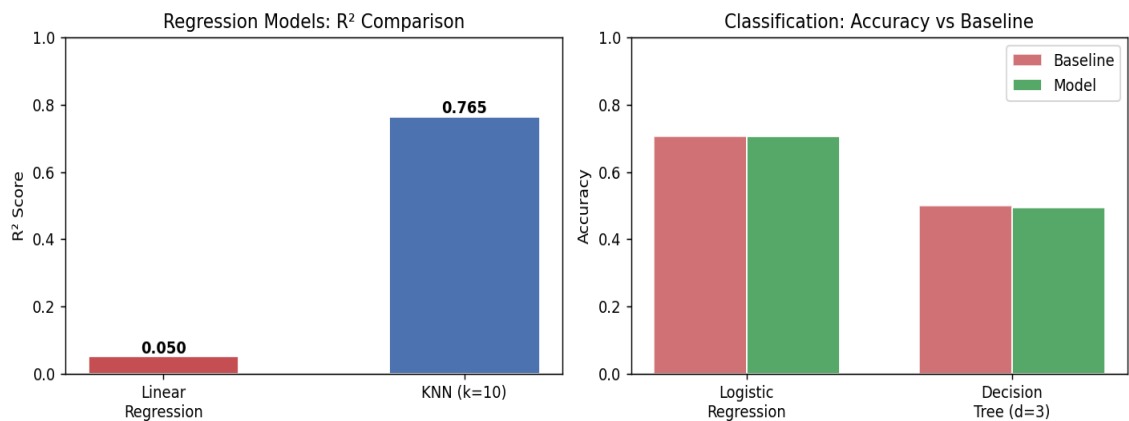


Figure 8: Summary — Regression R² comparison (left) and Classification accuracy vs baseline (right)

Finding	Evidence
Weather has weak linear predictive power	Linear R ² = 0.05
Listening cannot predict rain	Logistic accuracy = majority baseline
Nonlinear model dramatically improves	KNN R ² = 0.77
Condition encoded in time, not listening	DT month importance = 53%

The ML analysis deepens the story from Phase 3. Listening duration is predictable — but primarily from when in the year a day falls and from weekly routines, not weather itself. The jump from linear ($R^2=0.05$) to KNN ($R^2=0.77$) shows that complex, nonlinear patterns exist between season, routine, and listening. Weather is not the primary driver.

Limitations & Future Work

No Spotify audio features (valence, energy) were available in this phase — including them could reveal richer mood-based weather interactions. Class imbalance (72% non-rainy days) makes binary classification inherently difficult. Future work could apply Random Forest and Gradient Boosting (ensemble methods), use track-level features, or explore time-series models (LSTM, ARIMA) to capture sequential listening habits.